



2015-2017 C-MAX ENERGI (PHEV)

2015-2018 C MAX HYBRID (FHEV)



EMERGENCY RESPONSE GUIDE

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FOREWORD

The emergency response procedures for the C-MAX HYBRID and C-MAX ENERGI vehicles are similar to those for traditional gasoline-powered vehicles with the addition of special considerations for the high voltage electric system components.

The C-MAX HYBRID and C-MAX ENERGI vehicles use a conventional gasoline engine in addition to an electric motor to power the vehicle.

- Gasoline is stored in a traditional fuel tank.
- Electricity is stored in a high voltage battery pack.

The combination of a gasoline engine and electric motor provides for improved performance, reduced emissions and improved fuel economy. The C-MAX HYBRID high voltage systems are self-contained, never needing to be plugged into an electrical outlet for recharging. The system incorporates a generator that recharges the high voltage battery during cruising and braking. The C-MAX ENERGI's high voltage system can be plugged into an electric outlet for recharging to optimize the vehicle's extended electrical range. The system also incorporates a generator that recharges the high voltage battery during cruising and braking.

The information in this guide will allow response to emergencies involving the C-MAX HYBRID and C-MAX ENERGI vehicles to be as safe as non-HYBRID vehicles.

The C-MAX HYBRID and C-MAX ENERGI vehicles have been designed with many safety features for your protection. These features help provide safe access to the vehicle under various conditions. However, when approaching a high voltage vehicle in a fire, rescue or recovery situation, always follow one industry standard rule:

***ALWAYS ASSUME THE VEHICLE'S
HIGH VOLTAGE SYSTEM IS POWERED UP !***

C-MAX ENERGI and HYBRID VEHICLE IDENTIFICATION

C-MAX HYBRID vehicles can easily be identified by the HYBRID badges located on the left and right front doors. There is a HYBRID nameplate on the right side of the liftgate that also includes the green leaf/blue highway HYBRID icon.



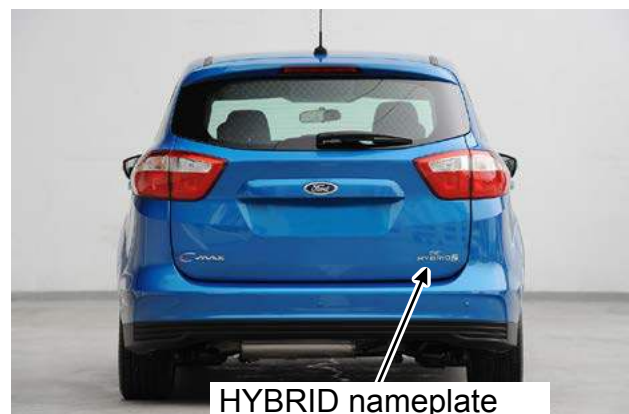
C-MAX ENERGI vehicles can easily be identified by the ENERGI badges located on the left and right front doors. There is an ENERGI nameplate on the left side of the liftgate that also includes the green leaf/blue highway HYBRID icon.



HYBRID/ENERGI
nameplate with HYBRID
icon on front doors.



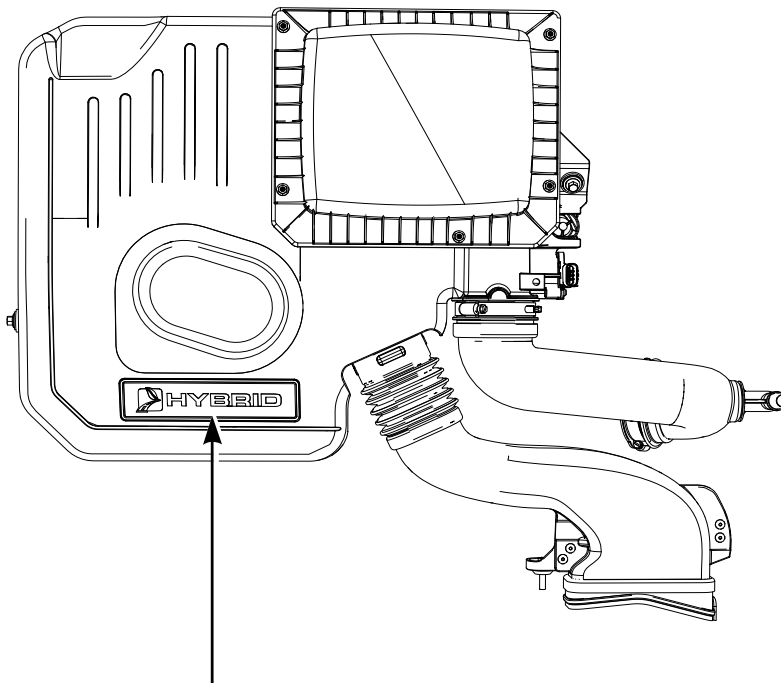
ENERGI nameplate
and icon on liftgate.



HYBRID nameplate
and icon on liftgate.

C-MAX ENERGI and HYBRID VEHICLE IDENTIFICATION (CONTINUED)

The C-MAX vehicle has a unique engine appearance cover with a raised HYBRID icon.



Unique HYBRID engine appearance cover
with raised lettering and icon

NOTE: C-MAX HYBRID shown, C-MAX ENERGI similar.

- The 5th, 6th and 7th digits of the Vehicle Identification Number (VIN) identify either the HYBRID or ENERGI vehicle lines.

VIN POSITIONS 5, 6, and 7	VEHICLE
P5A	HYBRID 5-door, SE
P5B	HYBRID 5-door, SEL
P5C	ENERGI 5-door, SEL (plug-in)

SAMPLE VIN: 3 F A D **P5A** 30AR100001

↑
C-MAX HYBRID FWD

HIGH VOLTAGE ELECTRICAL DISCONNECT FEATURES



WARNING: ELECTRIC VEHICLES DAMAGED BY A CRASH MAY HAVE COMPROMISED HIGH VOLTAGE SAFETY SYSTEMS AND PRESENT A POTENTIAL HIGH VOLTAGE ELECTRICAL SHOCK HAZARD. EXERCISE CAUTION AND WEAR APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT (PPE) SAFETY GEAR, INCLUDING HIGH VOLTAGE SAFETY GLOVES AND BOOTS. REMOVE ALL METALLIC JEWELRY, INCLUDING WATCHES AND RINGS. ISOLATE THE HIGH VOLTAGE SYSTEM AS DIRECTED BY THE FORD EMERGENCY RESPONSE GUIDE FOR THE VEHICLE. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH.

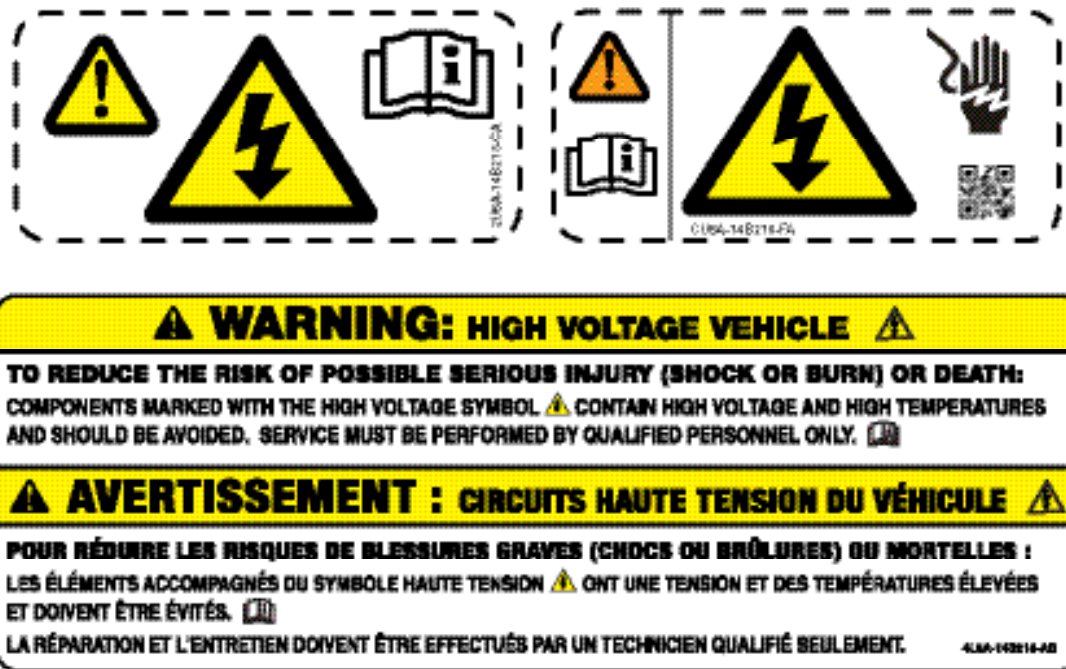
- If the vehicle is equipped with remote start, high voltage may be present in the system even when the ignition is off.
- ENERGI vehicles when plugged in for charging may have high voltage present in the system even when the ignition is off.

The following features have been incorporated into the C-MAX HYBRID and C-MAX ENERGI vehicles to allow for either simple or automatic shut off of the high voltage electrical systems.

- High voltage Fuse — In the event of a high voltage short circuit, the high voltage fuse opens, disabling the high voltage system.
- High voltage Interlock Circuit — Whenever a high voltage connector is disconnected, the high voltage interlock circuit opens and disables the high voltage system.
- Ignition Key/Button in the OFF Position — Any time the ignition is in the OFF position, the high voltage system is disabled.
- Battery High voltage Service Disconnect — Whenever the battery high voltage service disconnect (located behind the rear seat fold) is removed, the high voltage system is disabled. For additional information, refer to the battery high voltage service disconnect section of this manual.
- Thermal Sensors — In the event the battery is exposed to extremely hot ambient conditions and/or is being driven aggressively with compromised cooling, power limits will be employed to prevent overheating. However, if the battery is in use and becomes too hot, contacts will open and the vehicle will shut down. In some instances, if the ignition key is left in the ON position, this could cause the high voltage battery temperature to exceed 60°C (140°F). If this temperature is reached, the thermal sensors located near the high voltage battery automatically disable the high voltage battery.

HIGH VOLTAGE WARNING DECALS

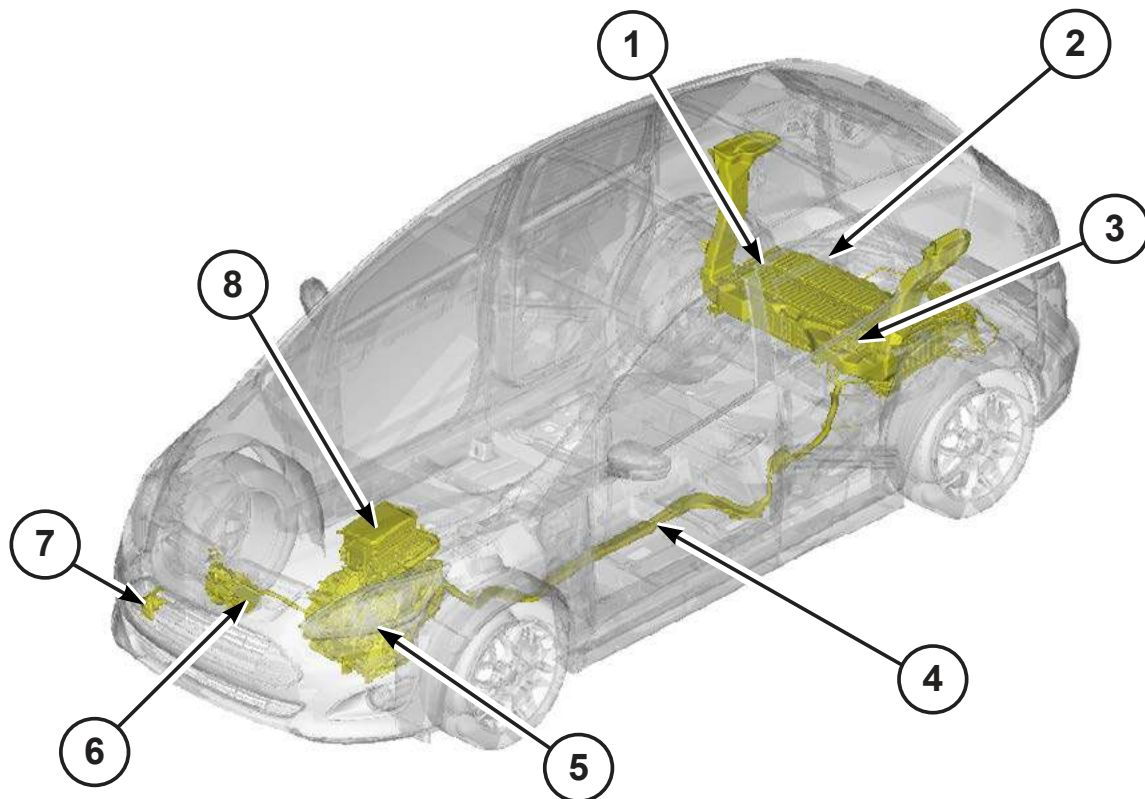
- On HYBRID vehicles, WARNING decals are located on components in the high voltage system throughout the vehicle. See the example below.



HYBRID COMPONENT LOCATION AND IDENTIFICATION

The following illustration provides the location, description and basic function of the HYBRID system components.

NOTE: All high voltage wires and harnesses are wrapped in orange insulation.



HYBRID COMPONENT LOCATION AND IDENTIFICATION (CONTINUED)

	COMPONENT	LOCATION / DESCRIPTION	FUNCTION
1	High voltage battery approximately 300 Volts DC	Located in the cargo area, behind the rear seat. Sealed lithium battery.	Provides high voltage storage for the vehicle's electric motor.
2	12-Volt battery (hidden from graphic)	Located in the cargo area. Typical automotive 6-cell lead/acid design.	Provides 12-volts for vehicle accessories. Receives current from the DC/DC converter (PIA to the high voltage battery) via a pre-fuse box in the engine compartment which runs to the 12V positive post.
3	High voltage service disconnect	Located behind the rear seat backrest on the driver side. Orange rectangle with a molded plastic handle, about 100 mm (4 in) in length.	Provides means to disconnect the high voltage battery for safely servicing vehicle.
4	High voltage wiring	Runs along the vehicle's floorpan from the high voltage battery to the HYBRID Powertrain Control Module (HPCM). The electric A/C compressor connects directly to the HPCM. Separate connections from the HPCM are made to the Electronically Controlled Continuously Variable Transaxle (ECVT).	Provides the physical connection between the high voltage battery and the vehicle's high voltage equipment.
5	Electronically Controlled Continuously Variable Transaxle (ECVT)	Transverse-mounted design. Contains the traction motor, generator motor and HYBRID electronics.	Provides rotational force to the wheels for vehicle propulsion and transmits rotational force to the engine's generator to recharge batteries.
6	Electric A/C compressor	Located in front of the engine. It has an orange high voltage wire attached to it.	Provides electric A/C operation; replaces the belt driven A/C compressor.

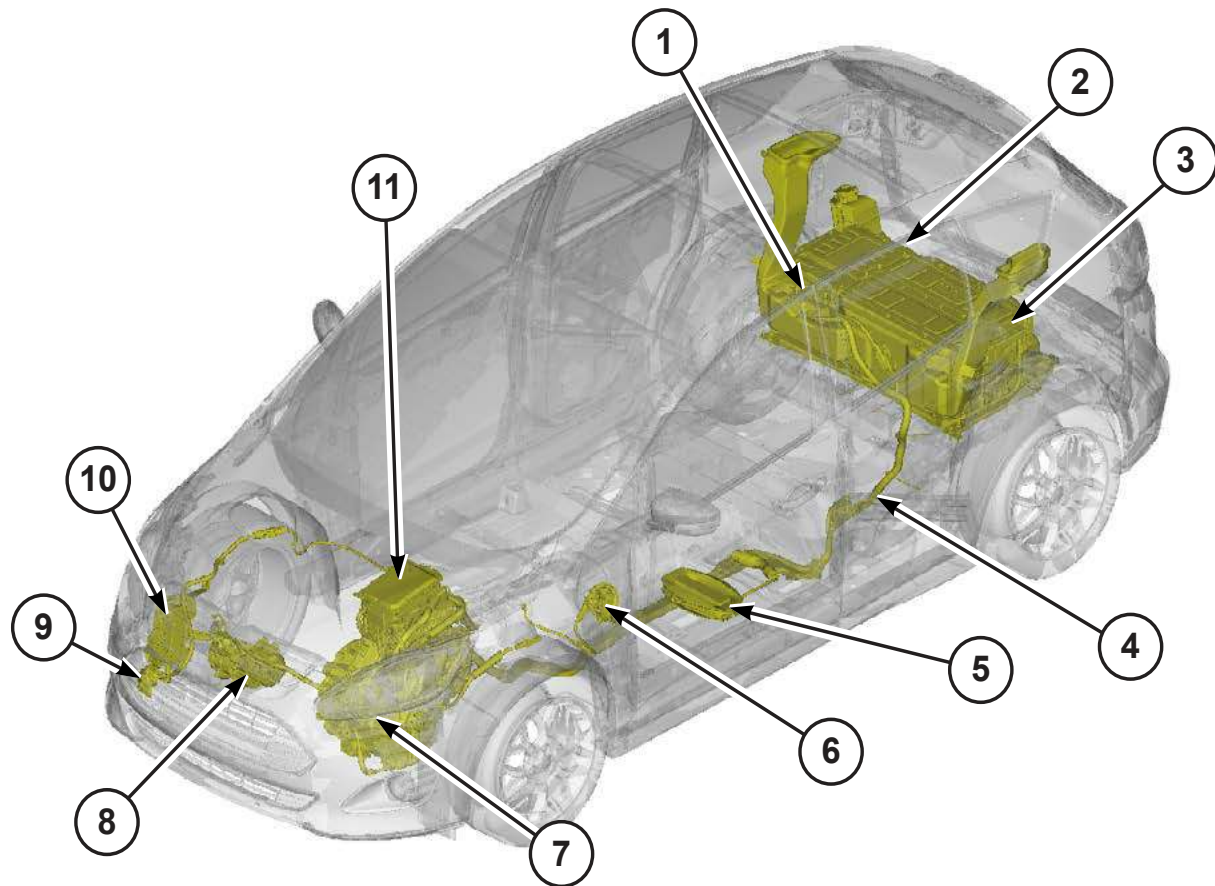
HYBRID COMPONENT LOCATION AND IDENTIFICATION (CONTINUED)

	COMPONENT	LOCATION / DESCRIPTION	FUNCTION
7	Electric water pump	Located in lower front of the engine. It is an electric 12-volt pump.	Provides water pump operation for the engine's cooling system. Replaces the belt driven water pump.
8	Transmission Control Module (TCM)	Located in front of the engine. Also referred to as the HPCM. Contains controlling electronics for managing propulsion.	Provides rotational force during Electrical Drive Modes. Generates electricity for recharging the high voltage battery.

ENERGI COMPONENT LOCATION AND IDENTIFICATION

The following illustration provides the location, description and basic function of the C-MAX ENERGI's system components.

NOTE: All high voltage wires and harnesses are wrapped in orange insulation.



ENERGI COMPONENT LOCATION AND IDENTIFICATION (CONTINUED)

	COMPONENT	LOCATION / DESCRIPTION	FUNCTION
1	High voltage service disconnect	Located behind the rear seat backrest on the driver side. Orange rectangle with a molded plastic handle, about 100 mm (4 in) in length.	Provides means to disconnect the high voltage battery for safely servicing the vehicle.
2	12-Volt battery (hidden from graphic)	Located in the cargo area. Typical automotive 6-cell lead/acid design.	Provides 12-volts for vehicle accessories. Receives current from the DC/DC converter (PIA to the high voltage battery) via a pre-fuse box in the engine compartment which runs to the 12V positive post.
3	High voltage battery approximately 300 Volts DC	Located in the cargo area, behind the rear seat. Sealed lithium battery.	Provides high voltage storage for the vehicle's electric motor.
4	High voltage wiring	Runs along the vehicle's floorpan from the high voltage battery to the HPCM. The electric A/C compressor connects directly to the HPCM. Separate connections from the HPCM are made to the Electronically Controlled Continuously Variable Transaxle (ECVT).	Provides the physical connection between the high voltage battery and the vehicle's high voltage equipment.
5	110V convenience cord	Located under the front driver side seat.	Provides 110V charging option to the high voltage battery when the vehicle is not in operation.
6	Charging port	Located on the driver side fender. It connects to the high voltage wiring in the engine compartment. It has an orange high voltage wire attached to it using an interlock connector.	Provides 110V or 220V charging options from a Level II charging station to the high voltage battery when the vehicle is not in operation.

ENERGI COMPONENT LOCATION AND IDENTIFICATION (CONTINUED)

	COMPONENT	LOCATION / DESCRIPTION	FUNCTION
7	Electronically Controlled Continuously Variable Transaxle (ECVT)	Transverse-mounted design. Contains the traction motor, generator motor and HYBRID electronics.	Provides rotational force to the wheels for vehicle propulsion and transmits rotational force to the engine's generator to recharge batteries.
8	Electric A/C compressor	Located in front of the engine. It has an orange high voltage wire attached to it.	Provides electric A/C operation; replaces the belt driven A/C compressor.
9	Electric water pump	Located in lower front of the engine. It is an electric 12-volt pump.	Provides water pump operation for the engine's cooling system. Replaces the belt driven water pump.
10	Electric heater	Located in the engine compartment on the passenger side.	Provides passenger compartment with heat without the need to turn the engine on.
11	Transmission Control Module (TCM)	Located in front of the engine. Also referred to as the HPCM. Contains controlling electronics for managing propulsion.	Provides rotational force during Electrical Drive Modes. Generates electricity for recharging the high voltage battery.

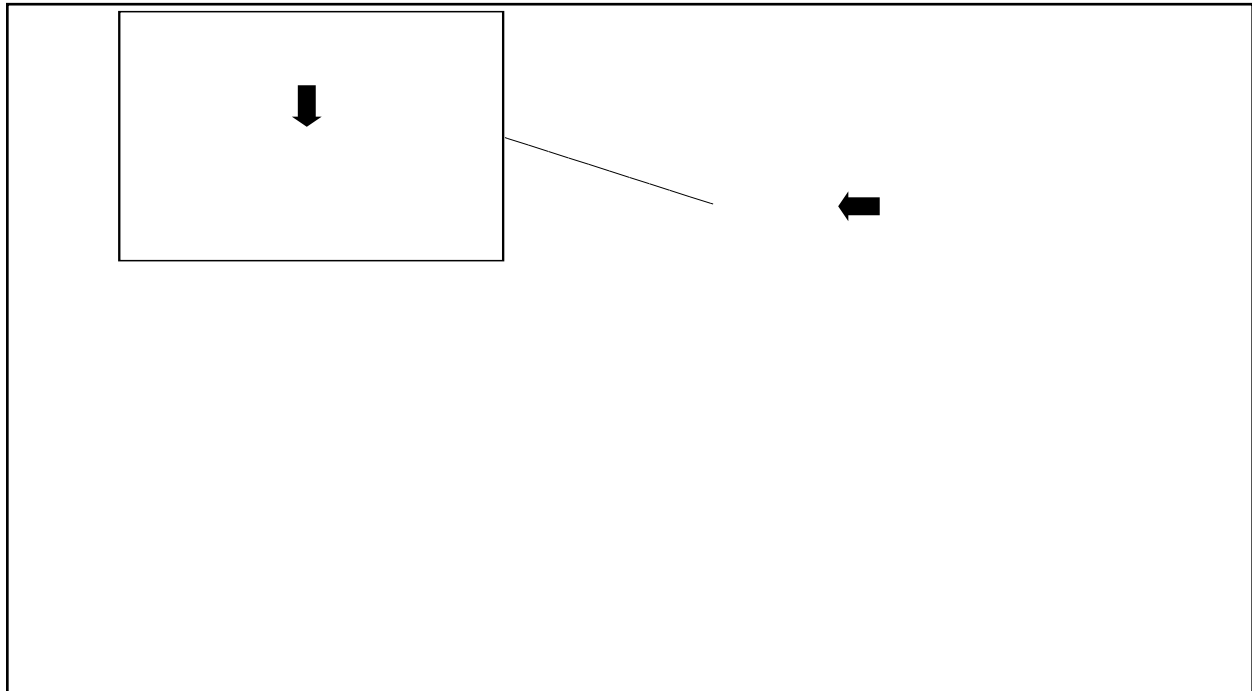
HIGH VOLTAGE SERVICE DISCONNECT - C-MAX HYBRID



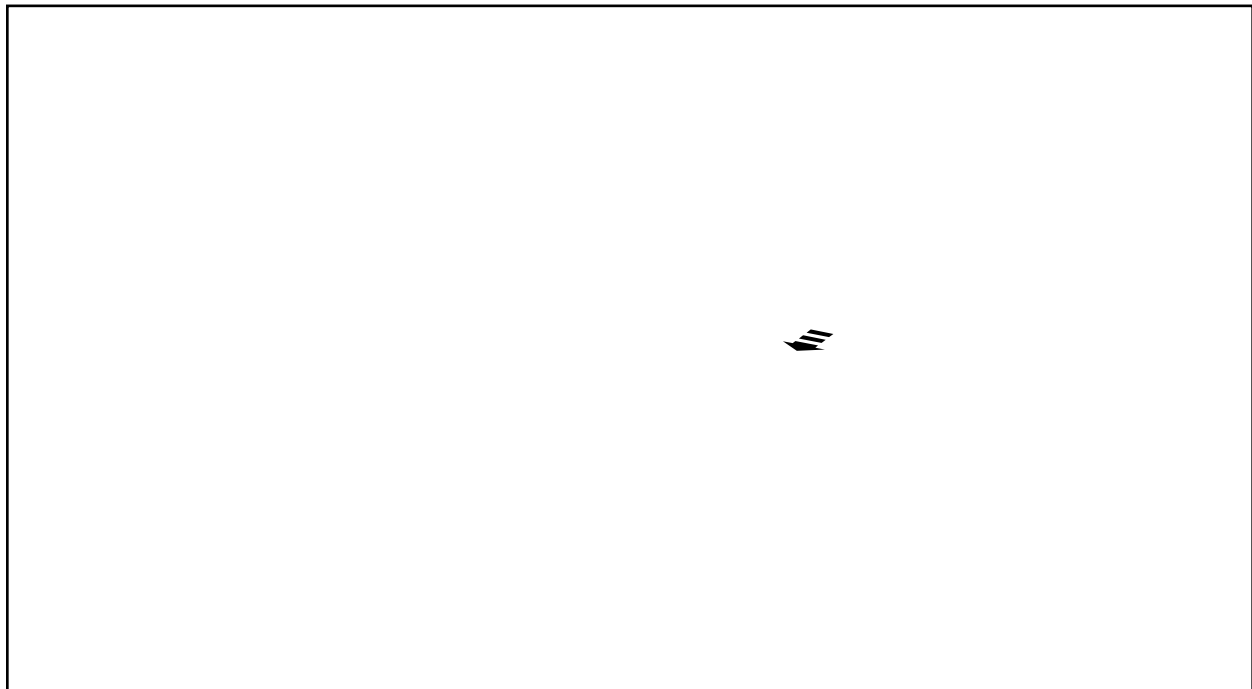
WARNING: TO PREVENT THE RISK OF HIGH VOLTAGE SHOCK, ALWAYS FOLLOW PRECISELY ALL WARNINGS AND SERVICE INSTRUCTIONS, INCLUDING INSTRUCTIONS TO DEPOWER THE SYSTEM. THE HIGH VOLTAGE SYSTEM UTILIZES APPROXIMATELY 300 VOLTS DC, PROVIDED THROUGH HIGH VOLTAGE CABLES TO ITS COMPONENTS AND MODULES. THE HIGH VOLTAGE CABLES AND WIRING ARE IDENTIFIED BY ORANGE HARNESS TAPE OR ORANGE WIRE COVERING. ALL HIGH VOLTAGE COMPONENTS ARE MARKED WITH HIGH VOLTAGE WARNING LABELS WITH A HIGH VOLTAGE SYMBOL. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH.

HIGH VOLTAGE SERVICE DISCONNECT - C-MAX HYBRID

1. Fold the rear seat backrest down, position the load floor up and remove the high voltage battery disconnect cover.



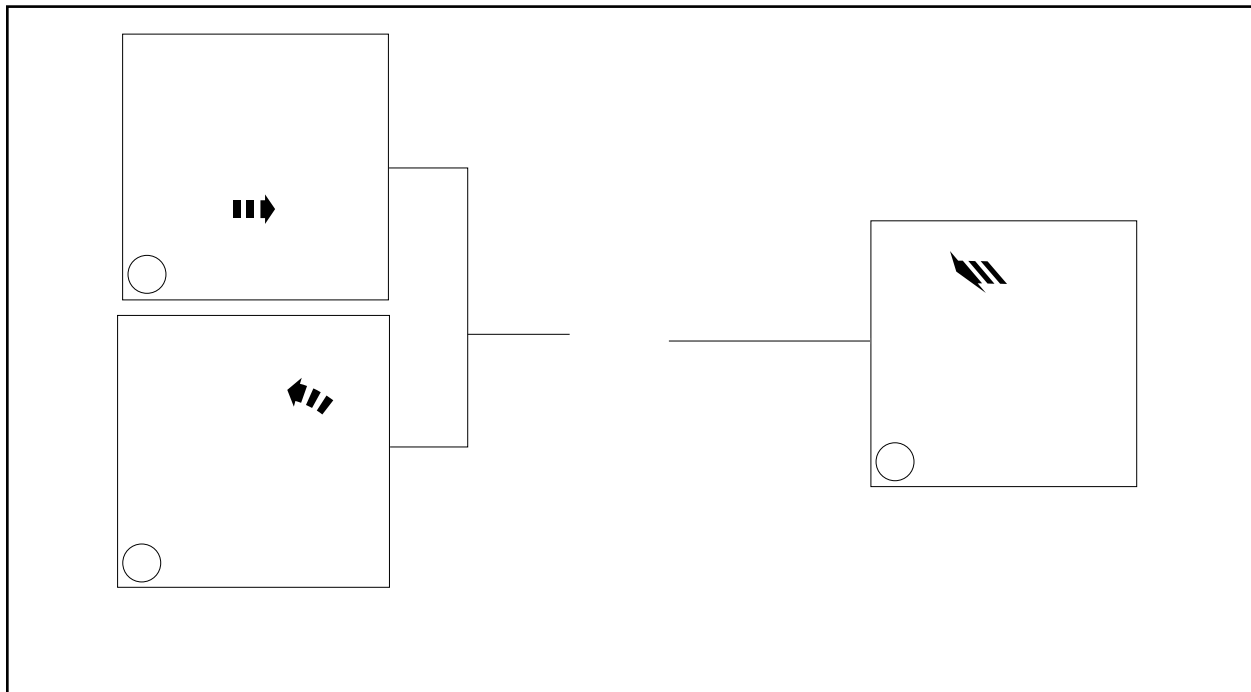
2. Pull to remove the battery high voltage service disconnect interlock cover.



HIGH VOLTAGE SERVICE DISCONNECT - C-MAX HYBRID (CONTINUED)

3. Remove the high voltage service disconnect:

1. Pull the Lever Position Assurance (LPA) tab toward the front of vehicle to remove.
2. Pull the lever handle outward to disengage the interlock.
3. Rotate the lever to horizontal and remove the battery high voltage service disconnect.



⚠ WARNING: THE HIGH VOLTAGE SYSTEM MAY RETAIN A DANGEROUS LEVEL OF VOLTAGE FOR A SHORT TIME AFTER THE SERVICE DISCONNECT HAS BEEN REMOVED. WAIT 5 MINUTES FOR THE VOLTAGE TO DISSIPATE BEFORE BEGINNING SERVICE. FAILURE TO FOLLOW THIS INSTRUCTION MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH.

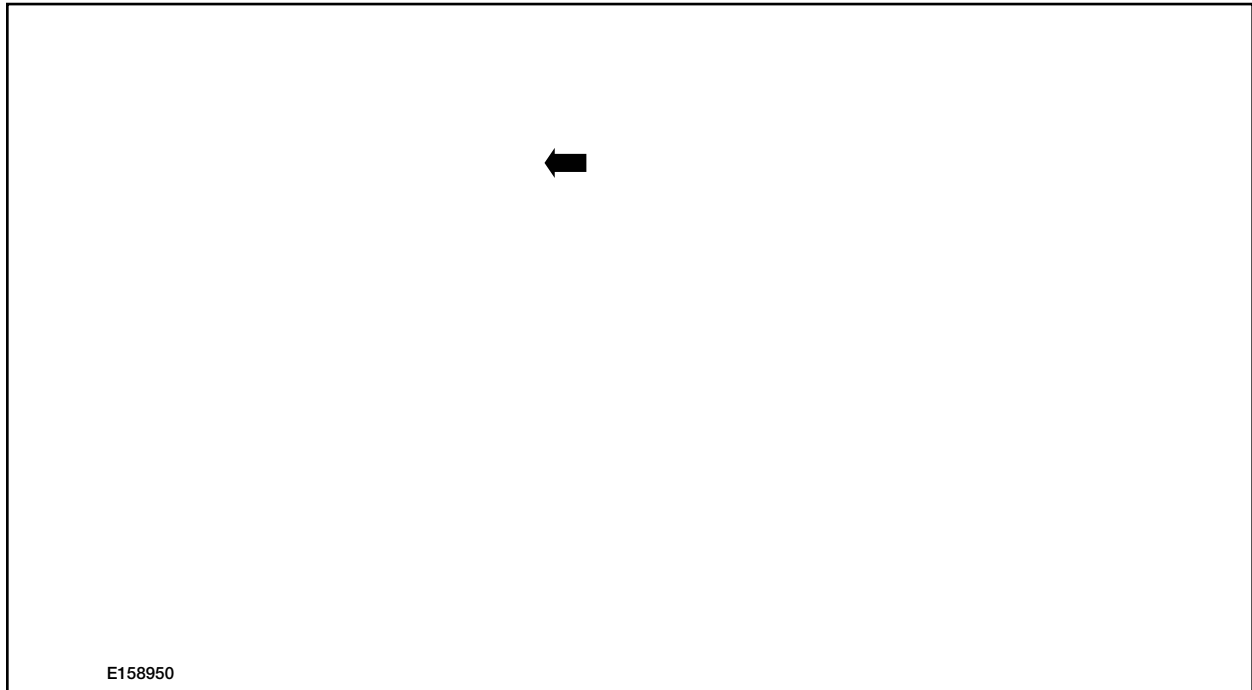
⚠ WARNING: REMOVING THE BATTERY HIGH VOLTAGE SERVICE DISCONNECT DOES NOT DISSIPATE VOLTAGE INSIDE THE BATTERY PACK. THE BATTERY PACK REMAINS LIVE AND DANGEROUS. CONTACT WITH THE HIGH VOLTAGE BATTERY PACK INTERNALS MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH.

HIGH VOLTAGE SERVICE DISCONNECT - C-MAX ENERGI

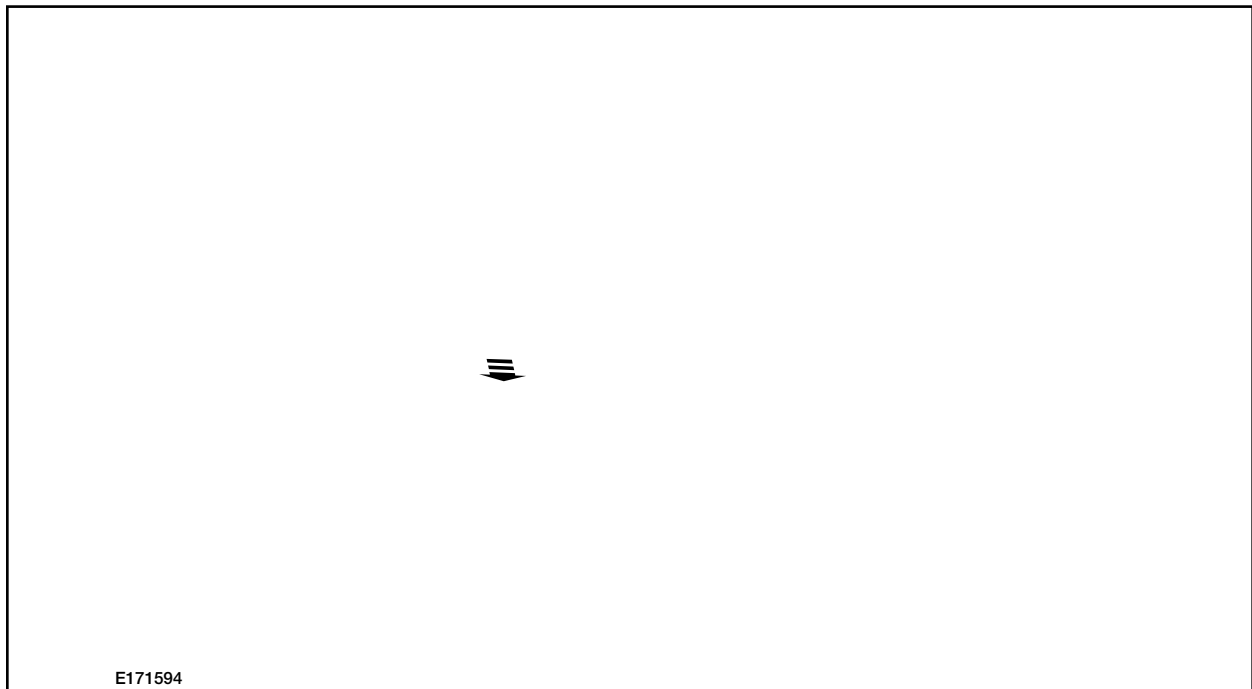
⚠ WARNING: TO PREVENT THE RISK OF HIGH VOLTAGE SHOCK, ALWAYS FOLLOW PRECISELY ALL WARNINGS AND SERVICE INSTRUCTIONS, INCLUDING INSTRUCTIONS TO DEPOWER THE SYSTEM. THE HIGH VOLTAGE SYSTEM UTILIZES APPROXIMATELY 300 VOLTS DC, PROVIDED THROUGH HIGH VOLTAGE CABLES TO ITS COMPONENTS AND MODULES. THE HIGH VOLTAGE CABLES AND WIRING ARE IDENTIFIED BY ORANGE HARNESS TAPE OR ORANGE WIRE COVERING. ALL HIGH VOLTAGE COMPONENTS ARE MARKED WITH HIGH VOLTAGE WARNING LABELS WITH A HIGH VOLTAGE SYMBOL. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH.

HIGH VOLTAGE SERVICE DISCONNECT - C-MAX ENERGI (CONTINUED)

1. Fold the rear backrest down and remove the battery high voltage battery disconnect cover.

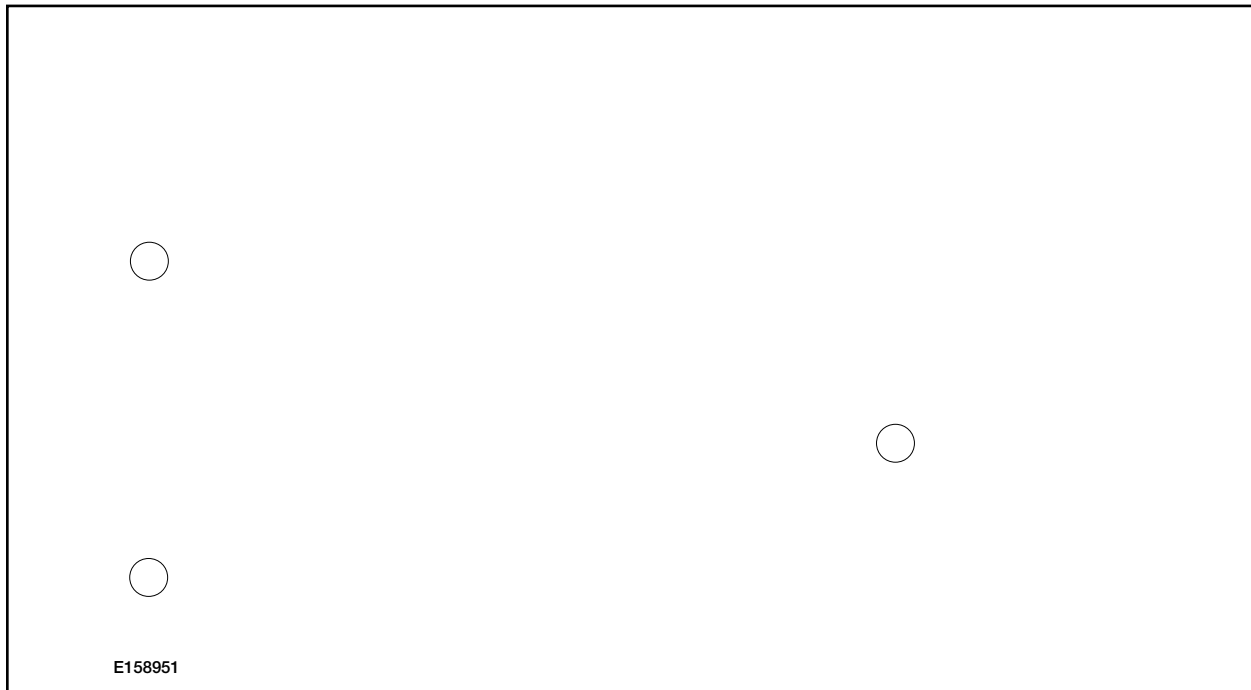


2. Pull to remove the battery high voltage service disconnect interlock cover.



HIGH VOLTAGE SERVICE DISCONNECT - C-MAX ENERGI (CONTINUED)

3. Remove the battery high voltage service disconnect:
 1. Pull the Lever Position Assurance (LPA) tab toward the front of vehicle to remove.
 2. Pull the lever handle outward to disengage the interlock.
 3. Rotate the lever to horizontal and remove the battery high voltage service disconnect.



 **WARNING: THE HIGH VOLTAGE SYSTEM MAY RETAIN A DANGEROUS LEVEL OF VOLTAGE FOR A SHORT TIME AFTER THE SERVICE DISCONNECT HAS BEEN REMOVED. WAIT 5 MINUTES FOR THE VOLTAGE TO DISSIPATE BEFORE BEGINNING SERVICE. FAILURE TO FOLLOW THIS INSTRUCTION MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH.**

 **WARNING: REMOVING THE BATTERY HIGH VOLTAGE SERVICE DISCONNECT DOES NOT DISSIPATE VOLTAGE INSIDE THE BATTERY PACK. THE BATTERY PACK REMAINS LIVE AND DANGEROUS. CONTACT WITH THE HIGH VOLTAGE BATTERY PACK INTERNALS MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH.**

HIGH VOLTAGE BATTERY PACK

 **WARNING: ELECTRIC VEHICLES DAMAGED BY A CRASH MAY HAVE COMPROMISED HIGH VOLTAGE SAFETY SYSTEMS AND PRESENT A POTENTIAL HIGH VOLTAGE ELECTRICAL SHOCK HAZARD. EXERCISE CAUTION AND WEAR APPROPRIATE PERSONAL PPE SAFETY GEAR, INCLUDING HIGH VOLTAGE SAFETY GLOVES AND BOOTS. REMOVE ALL METALLIC JEWELRY, INCLUDING WATCHES AND RINGS. ISOLATE THE HIGH VOLTAGE SYSTEM AS DIRECTED BY THE FORD EMERGENCY RESPONSE GUIDE FOR THE VEHICLE. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH.**

HIGH VOLTAGE BATTERY PACK - (CONTINUED)



WARNING: FIRES IN CRASH-DAMAGED ELECTRIC VEHICLES MAY EMIT TOXIC OR COMBUSTIBLE GASES. SMALL AMOUNTS OF EYE, SKIN OR LUNG IRRITANTS MAY BE PRESENT. WEAR PERSONAL PPE AND SELF-CONTAINED BREATHING APPARATUS WHEN WORKING IN CLOSE PROXIMITY OR IN A CONFINED AREA, SUCH AS A TUNNEL OR GARAGE. VENTILATE THE VEHICLE INTERIOR BY OPENING VEHICLE WINDOWS OR DOORS. VENTILATE THE WORKING AREA. FAILURE TO FOLLOW THIS INSTRUCTION MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH.

- Do not cut the high voltage battery case. Do not penetrate the batteries or case in any way.
- The high voltage battery is located in the forward portion of the luggage compartment. The high voltage battery lies flat just behind the rear seat backrest under the luggage compartment cargo cover.
- Removing the battery high voltage service disconnect will disconnect the high voltage from the vehicle. The battery pack will remain live and dangerous.
- The total voltage of the battery pack is approximately 300 volts DC.
- The battery cells contain a base electrolyte, consisting of lithium hexafluorophosphate and organic solvents as the dominant active ingredient, that is absorbed in special polymeric film. The electrolyte will not leak from the battery under most conditions. However, if the battery is crushed, it is possible for a small amount of electrolyte to leak.



CAUTION: Observe the following when working with a damaged battery:

- If possible, isolate and avoid contact with high voltage vehicle components. If contact with the high voltage system cannot be avoided, PPE such as splash shield or safety goggles, gloves (latex, rubber or Nitrile), an apron or overcoat, and rubber boots are required when handling damaged batteries. Exposure to electrolyte could cause skin and/or eye irritation/burns. If exposed, rinse with large amounts of water for 10-15 minutes.
- If the battery is exposed to intense heat (or other abusive conditions), it is possible that flammable gases and liquid (electrolyte) have been released from the cells. The gases include combustible hydrocarbons such as methane, toxic gases such as carbon monoxide, and very small amounts of eye/skin/lung irritants such as hydrofluoric acid. Appropriate precautions should be taken to make sure the area is properly ventilated; such as opening the vehicle's windows or doors. First responders should wear PPE and self-contained breathing apparatus to safeguard against thermal, electrical, respiratory and skin/eye hazards.

APPROACHING A DAMAGED HIGH VOLTAGE VEHICLE



WARNING: ELECTRIC VEHICLES DAMAGED BY A CRASH MAY HAVE COMPROMISED HIGH VOLTAGE SAFETY SYSTEMS AND PRESENT A POTENTIAL HIGH VOLTAGE ELECTRICAL SHOCK HAZARD. EXERCISE CAUTION AND WEAR APPROPRIATE PPE SAFETY GEAR, INCLUDING HIGH VOLTAGE SAFETY GLOVES AND BOOTS. REMOVE ALL METALLIC JEWELRY, INCLUDING WATCHES AND RINGS. ISOLATE THE HIGH VOLTAGE SYSTEM AS DIRECTED BY THE FORD EMERGENCY RESPONSE GUIDE FOR THE VEHICLE. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH.



WARNING: DAMAGED ELECTRIC VEHICLES SUBMERGED IN WATER PRESENT A POTENTIAL HIGH VOLTAGE ELECTRICAL SHOCK HAZARD. EXERCISE CAUTION AND WEAR APPROPRIATE PERSONAL PPE SAFETY GEAR, INCLUDING HIGH VOLTAGE SAFETY GLOVES AND BOOTS. REMOVE ALL METALLIC JEWELRY, INCLUDING WATCHES AND RINGS. DO NOT ATTEMPT TO EXTRACT THE VEHICLE UNTIL THE HIGH VOLTAGE BATTERY HAS DISCHARGED AS INDICATED BY THE ABSENCE OF BUBBLING OR FIZZING. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH.

1. FOLLOW EXISTING TRAINING AND INCIDENT COMMANDER DIRECTION

- Emergency responders should use **LARGE** amounts of water if fire is present or suspected and, keeping in mind that fire can occur for a considerable period after the crash, should proceed accordingly.
- This guide provides only supplemental information as it pertains to the C-MAX HYBRID and C-MAX ENERGI vehicles. The same rules apply when approaching any potential high voltage situation. Always follow your high voltage safety training. Some precautions to take in a high voltage situation include:
 - Remove all jewelry: Watches, necklaces and earrings. Remove any metal objects that are conductors of electricity.
 - Wear the necessary PPE's such as high voltage rubber gloves, face shield, insulated boots, protective raincoat and apron.
- Bring the following equipment:
 - Class ABC powder-type fire extinguisher.
 - A non-conductive object, about 1.5 m (5 ft) long. This can be used to safely push someone away from the vehicle if they accidentally come in contact with high voltage.

APPROACHING A DAMAGED HIGH VOLTAGE VEHICLE (CONTINUED)

2. APPROACHING A DAMAGED VEHICLE

- Disable the high voltage electrical system using as many of the following steps as possible:
 - Secure the vehicle: Place the shift lever into the PARK position. Place the ignition in the OFF position to disconnect the high voltage system. Block the wheels, if necessary.
 - If possible, remove the battery high voltage service disconnect(s). For additional information, refer to the high voltage service disconnect section in this guide. If the service disconnect(s) cannot be removed, avoid the high voltage system components and wear appropriate PPE as outlined in this guide.
- If the vehicle is on fire, use a Class ABC powder-type extinguisher to contain and smother the flames. If water is being used, **LARGE** amounts of water will be required to extinguish the flames. A fire-hydrant or dedicated fire hose can supply the needed amount. Water can cause some degree of arcing/shorting across the cell and/or battery terminals; it can also react with the electrolyte from the cells to generate additional combustible gas and other byproducts such as hydrofluoric acid. However, the cooling and smothering effects of flushing the affected article with large amounts of water and/or other fire suppression material is still beneficial for minimizing the severity of the event.
- If the vehicle has any exposed cables, make sure to wear high voltage rubber gloves and other appropriate PPE. Do not touch any broken or damaged high voltage orange cables. Treat severed cables as if they contain high voltage.
- If the vehicle is submerged in water, varying degrees of arcing/shorting within the battery will take place. Do not touch any high voltage components or orange cables while extricating the occupant(s). Do not remove the vehicle until you are sure the high voltage battery is completely discharged. A submerged high voltage battery may produce a fizzing or bubbling reaction to the water. If fizzing or bubbling is observed, the high voltage battery will be discharged when the fizzing or bubbling has completely stopped. However, you should still treat it as if it was not discharged.

APPROACHING A DAMAGED HIGH VOLTAGE VEHICLE (CONTINUED)

3. IF THE HIGH VOLTAGE BATTERY CASE HAS BEEN RUPTURED

- Use a fire hose to hose the area down with **LARGE** amounts of water just like you would any other battery.

4. MOVING DAMAGED VEHICLES - WRECKER DRIVERS

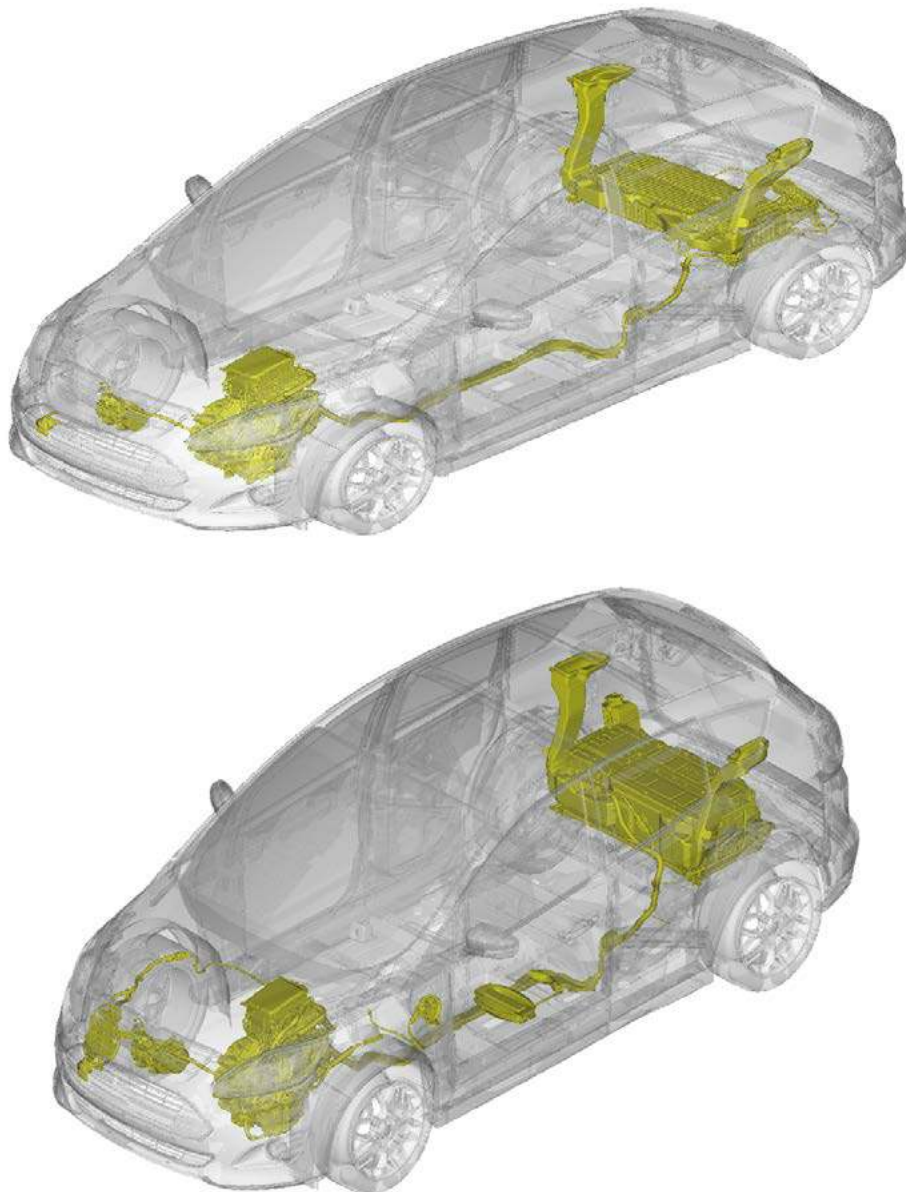
NOTICE: Never attempt to tow the vehicle with its drive wheels on the ground.

- Rather than attempt to discharge a propulsion battery, an emergency responder, tow truck operator, or storage facility manager should contact the vehicle manufacturer. If possible, remove the high voltage service disconnect. For additional information, refer to the High voltage Service Disconnect section in this guide.
- Be alert. There is potential for delayed fire with damaged lithium-ion batteries.
- Call an authorized Ford Dealer or vehicle manufacturer representative, if necessary, to determine additional steps that you should take to safely recover or transport the vehicle.
- Follow the guidelines in the Wrecker Towing Manual.
 - Front Tow: Wheel Lift (FWD)
 - Flatbed: FWD
- If a vehicle with a high voltage battery is to be scrapped, the high voltage battery must be disposed of following all local, state/provincial and federal guidelines.
- Operators of tow trucks and vehicle storage facilities should make sure the damaged vehicle is kept in an open area instead of inside a garage or other enclosed building.

APPROACHING A DAMAGED HIGH VOLTAGE VEHICLE (CONTINUED)

5. HIGH VOLTAGE SYSTEM - DO NOT CUT ZONES

- If possible, remove the battery high voltage battery service disconnect before attempting any extrication procedure. Always assume the high voltage cabling and components are powered up.
- If occupant extrication is necessary, always use caution when cutting near the vehicle high voltage system components. Do not cut any of the high voltage under-vehicle or under-hood cabling (all high voltage cabling is orange). High voltage cabling runs from the high voltage batteries under the left hand side of the vehicle to the underhood compartment. Refer to the diagram below for the no cut zones.



Damaged Vehicle Guidance for Ford Motor Company Electric and HYBRID Electric Vehicles Equipped with High voltage Batteries (Towing and Recovery Operators and Vehicle Storage Facilities)

Call an authorized service center or vehicle manufacturer representative, if necessary, to determine additional steps that you should take to safely recover or transport the vehicle.

In the event of damage or fire involving an Electric Vehicle (EV) or HYBRID Electric Vehicle (HEV):

- Always assume the high voltage battery and associated components are energized and fully charged.
- Exposed electrical components, wires and high voltage batteries present potential high voltage shock hazards.
- Venting/off-gassing high voltage battery vapors are potentially toxic and flammable.
- Physical damage to the vehicle or high voltage battery may result in immediate or delayed release of toxic and/or flammable gases and fire.

Vehicle Shutdown and High voltage System Disabling Identify Vehicle

- Determine if the vehicle is an electric or HYBRID-electric vehicle, and if it is, advise your dispatch and all other responders that an electric or HYBRID-electric vehicle is involved.
- To identify potential symptoms of a damaged high voltage system, contact an authorized service center or vehicle manufacturer representative. Refer to the Vehicle Owner Manual, Emergency Placard (included in the Vehicle Owner Manual) and/or the Emergency Response Guide for appropriate contact information.
- If you detect leaking fluids, sparks, smoke, flames, increased temperature, gurgling, popping or hissing noises from the high voltage battery compartment, ventilate the passenger area such as roll down windows or open doors) and call 911.
- Be alert. There is a potential for delayed fire with damaged lithium-ion batteries.

Recovering/Transporting Vehicle

NOTICE: Never attempt to tow the vehicle with its drive wheels on the ground.

Recovering/Transporting Vehicle (CONTINUED)

- Call an authorized service center or vehicle manufacturer representative, if necessary, to determine additional steps that should be taken to safely recover or transport the vehicle.
- Always approach the vehicle from the sides to stay out of potential travel path. It may be difficult to determine if the vehicle is running due to lack of engine noise.
- Place vehicle into park (P), set the parking brake, turn off the vehicle, activate hazard lights, and remove keys to a distance at least 16 feet from the vehicle until loading the vehicle for transport.
- Refer to vehicle manual/recovery guide to locate proper attachment/connection points and transport method.
- Avoid contact with orange high voltage cabling and areas identified as high voltage risk by warning labels.

Storing Vehicle

- **For vehicles in the United States, notify Ford Motor Company 1-800-392-3673 (then follow the prompts on the voice response menu) or an authorized Ford Dealer as soon as possible as there may be additional steps necessary to secure, discharge, handle, and/or store the high voltage battery and vehicle.**
- **For vehicles in Canada, notify Ford Motor Company 1-800-565-3673 (then follow the prompts on the voice response menu), or an authorized Ford dealer as soon as possible as there may be additional steps necessary to secure, discharge, handle, and/or store the high voltage battery and vehicle.**
- Do not store a severely damaged vehicle with a lithium-ion battery inside a structure or within 50 feet of any structure or vehicle.
- Make sure that passenger and cargo compartments remain ventilated.
- Prior to placing, and while located in storage area/tow lot, continue to inspect vehicle for leaking fluids, sparks, smoke, flames, gurgling or bubbling sounds from the high voltage battery and call 911 if any of these are detected.
- Maintain clear access to stored vehicles for monitoring and emergency response if needed.

For specific information and safety preparation regarding the high voltage system, refer to WWW.MOTORCRAFTSERVICE.COM, select quick guides for the appropriate vehicle emergency response guide.

DAMAGED BATTERY ELECTRIC VEHICLE STORAGE PLACARD

- If the vehicle and/or battery high voltage system is damaged, place a sign indicating that it is a battery electric vehicle with potentially dangerous high voltage. See example below:

**WARNING:
BATTERY ELECTRIC VEHICLE WITH
POSSIBLE HIGH VOLTAGE SYSTEM
DAMAGE. DO NOT TOUCH!**

**WARNING:
BATTERY ELECTRIC VEHICLE WITH
POSSIBLE HIGH VOLTAGE SYSTEM
DAMAGE. DO NOT TOUCH!**

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